Consistency of Force Prediction in Single Execution Figure 5 displays the alignment between ForceFlow's predicted contact force curve (red) and the real measured force (blue) across six representative tasks. In short-contact tasks (Stamp/Plug/Press), during the contact establishment phase, the rising trend of the predicted force aligns highly with the real contact force. During the contact release phase (task completion or leaving the surface), the predicted force declines synchronously with the real force. This suggests that the model can infer the impending contact state via force history, vision, and proprioception. In continuous tasks (Clean Whiteboard/Clean Vase), the predicted force smoothly adjusts following surface changes, without exhibiting significant high-frequency oscillation or jitter. This indicates that the model is not simply regressing to a mean force value but has learned state-based continuous force regulation.

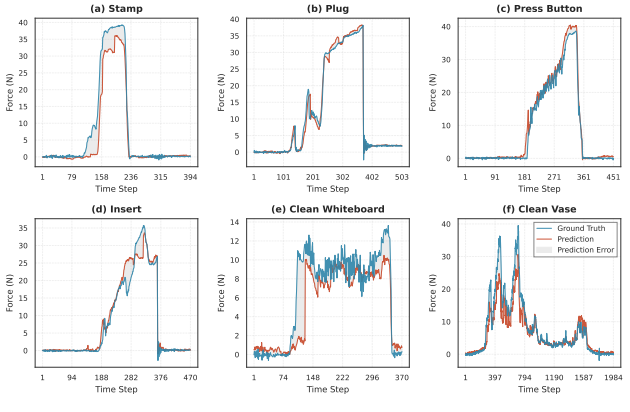

Figure 5. Comparison of Predicted and Measured Forces. Alignment between the predicted (red) and measured ground truth (blue) contact forces. Shaded areas represent prediction error, demonstrating high temporal fidelity during interaction.

4.4. RQ3: Generalization to OOD Environments

This section evaluates the robustness of ForceFlow across complex real-world variations by decoupling visual localization from force regulation. We analyze two critical dimensions of generalization: (1) **Force-side Generalization**, where contact objects or tools change but spatial locations remain fixed, and (2) **Vision-side Generalization**, where target objects are placed in OOD spatial locations.

Force-side Generalization: Zero-Shot Adaptability. We evaluate physical adaptability by replacing training tools with unseen variants without fine-tuning (see Figure 6a-b). We conducted 10 independent trials per task using objects with distinct physical properties: for **Clean Vase** and **Whiteboard**, we introduced two novel erasers with significantly different stiffness and thickness; for **Press Button**, we utilized a new bottle with a distinct height and spring trigger threshold.

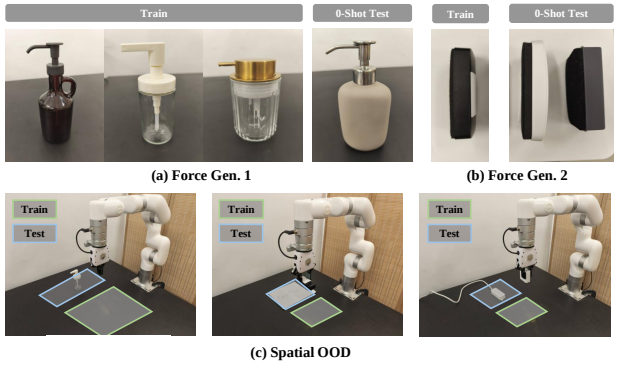

Figure 6. **OOD Evaluation Setup.** (a)-(b) **Force-side Generalization:** Unseen objects (e.g., distinct bottles, different erasers) used for zero-shot testing. (c) **Vision-side Generalization:** The testing workspace (marked "Test") is spatially disjoint from the training distribution (marked "Train"), requiring V2F for localization.

Table 3. **Zero-Shot Force Generalization (SR %).**

Method	Press	Clean WB	Clean Vase
$\pi_{0.5}$	0%	0%	0%
ACT	0%	0%	0%
Diffusion Policy	0%	0%	0%
ForceVLA	40%	90%	0%
ForceFlow (Ours)	**80%**	**100%**	**60%**

As shown in Table 3, vision-centric baselines fail entirely when tool friction or object elasticity changes, as they cannot perceive high-frequency physical feedback. In contrast, ForceFlow maintains high success rates by adaptively adjusting contact forces through its temporal force perception and active prediction mechanisms. By encoding F_{hist}, the model perceives contact states and achieves proactive compliance even when the physical interaction parameters deviate from the training distribution.

Vision-side Generalization: Hierarchical Grounding. To evaluate robustness against spatial shifts, we configure a testing workspace that is spatially disjoint from the training distribution, as illustrated in Figure 6(c). To tackle large spatial variations, we implement V2F, specifically utilizing Embodied-R1 (Yuan et al., 2025) as the high-level planner. This hierarchical strategy allows the system to localize targets in OOD positions significantly outside the training distribution.

Results in Table 4 highlight the necessity of V2F integration. Without semantic guidance, low-level policies (including standalone baselines) fail to locate objects in OOD regions. By utilizing V2F, the system predicts semantic pixel coordinates from the global view I_{fix} to guide initial movement. Once the end-effector enters the local interaction window, ForceFlow takes over for high-frequency regulation. This